

CASE REPORT



Scurvy presenting in a homeless 32-year-old male: a case report

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Vitamin C deficiency, otherwise known as scurvy, is one the oldest known diseases. Although its prevalence has substantially diminished, certain populations such as smokers, alcoholics, elderly, and those with malabsorptive syndromes remain at risk. Deficiency presents with perifollicular hemorrhage, corkscrew hairs, and petechiae that can evolve into ecchymoses and purpura. Diagnosis starts with clinical suspicion and is confirmed with plasma and leukocyte ascorbic acid levels. Early suspicion is key. Treatment with vitamin C offers an excellent prognosis and avoids unnecessary workup for differential diagnoses. In our case, a 45-year-old male alcoholic with progressively worsening symptoms was found to have scurvy. Treatment was initiated with vitamin C supplementation, and he had remarkable improvement in his symptoms within weeks. This case demonstrates that scurvy is not a disease of history and must be considered in modern medicine, especially as homelessness with concomitant poor nutrition continues to increase.

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INTRODUCTION

Scurvy is a clinically diagnosed condition that has been studied for centuries [1, 2]. Vitamin C deficiency clinically manifests with nonspecific malaise, fatigue, and lethargy that progress to systemic symptoms including anemia, myalgias, easy bruising, petechiae, gum disease, poor wound healing, and mood changes. Untreated cases can cause generalized edema, severe jaundice, hemolysis, acute spontaneous bleeding, neuropathy, fever, convulsions, and death [3]. Its incidence is low in modern times due to widespread availability of vitamin C-rich fruits and vegetables, but risk factors include poor access to these foods, mental health disorders, substance abuse, obesity, poor diets, and malabsorption syndromes [4].

CASE REPORT

A 45-year-old homeless male with a history of polysubstance use presented to the emergency department for a one-month history of gingival swelling and bleeding, bilateral lower extremity bruising, and edema. His symptoms progressively worsened, and he had a pre-syncope episode before presentation. The patient denied any recent trauma, hematochezia, melena, or epistaxis. He had no history of bleeding diathesis, or clotting disorders. Family history was negative for bleeding disorders. Prior to admission, he had been drinking 6–8 beers daily and had not consumed fresh fruit or vegetables for weeks. Vitals revealed tachycardia of 102 but were otherwise unremarkable. On physical exam, he was morbidly obese with an unkempt appearance. A dental exam revealed poor dentition and gingival bleeding as seen in Fig. 1. The patient had extensive bilateral upper (Fig. 2) and lower extremity ecchymosis most pronounced in bilateral thighs. There was no evidence of hepatosplenomegaly. Initial labs revealed low hemoglobin of 5.8 g/dL (ref 12.6–16.7 g/dL) with elevated platelet

count of 408 $10^3/\mu\text{L}$ (ref 139–361 $\times 10^3/\mu\text{L}$) and WBC 11,200/ μL (ref 4.4–10.5 $\times 10^3/\mu\text{L}$). The basic metabolic panel showed hyponatremia of 126 mmol/L (ref 136–145 mmol/L), bicarbonate level of 18 mmol/L (ref 21–31 mmol/L) with an anion gap of 14 (ref 2–12), and mildly decreased chloride 94 mmol/L (ref 98–107 mmol/L). EKG showed only sinus tachycardia.

The patient was immediately transfused 2 units of packed red blood cells, and hemoglobin improved to 8.1 g/dL. His sodium and acidosis resolved with fluids. Additional workup showed an indirect hyperbilirubinemia with a direct bilirubin 0.8 mg/dL (ref 0–0.2 mg/dL) and total bilirubin 3.4 mg/dL (ref 0.3–1.0 mg/dL). Iron studies revealed low iron level 39 $\mu\text{g/dL}$ (ref 50–212 $\mu\text{g/dL}$) and iron saturation 10% (ref 20–50%), with normal ferritin 78.5 ng/mL (ref 23.9–336.2 ng/mL) and TIBC 380 $\mu\text{g/dL}$ (ref 250–400 $\mu\text{g/dL}$). Hemolytic investigation revealed elevated haptoglobin 324 mg/dL (ref 44–215 mg/dL) and significant reticulocytosis 10.12% (ref 0.5–2.2%), with normal LDH. Peripheral smear was significant only for normocytic, normochromic anemia and hyperlobated neutrophils. PT/INR was not prolonged and fibrinogen levels were normal. B12 level was 196 pg/mL (ref 180–914 pg/mL) and folate was low at 4.3 ng/mL (ref 5.9–24.8 ng/mL). A computed tomography (CT) scan without contrast of the bilateral thighs revealed nonspecific subcutaneous stranding and possible edema without evidence of soft tissue injury.

Endoscopy was deferred. Bleeding disorders such as von Willebrand Disease and hemophilia were ruled out. Von Willebrand Factor (VWF) antigen and activity were within normal limits and factor VIII level was elevated at 395% (ref 55–200%). Paraproteinemia labs were unremarkable including serum protein electrophoresis (SPEP) with immunofixation and serum free light chains. Finally, a vitamin C level was found to be undetectable (<0.1), confirming the diagnosis of scurvy. The patient was given vitamin C 1000 mg PO BID in addition to folate and B12

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Fig. 1 Oral cavity examination. The initial physical examination showing gingivitis at red arrow.



Fig. 2 Inner thigh examination. Lower extremity with extensive ecchymosis most pronounced in the thighs (red arrow).

supplementation for his chronic alcohol abuse. Within two days of initiating vitamin C replacement, he began experiencing improvement in the bruising and swelling. Complete resolution of symptoms was seen in the weeks following his supplementation.

DISCUSSION

Ascorbic acid is a ketolactone synthesized from glucose. Humans are among a small group of species unable to oxidize one of the key intermediates along this pathway. Ascorbic acid plays numerous physiological roles, ranging from a key cofactor in collagen biosynthesis, to promoting nonheme iron absorption as an electron donor, and aiding in fatty acid transport, cholesterol metabolism, reactive oxygen species elimination, and dopamine conversion to norepinephrine [1, 2].

Clinical presentations are correlated to these physiological roles. Impaired collagen synthesis leads to the hallmark symptoms of follicular hyperkeratosis, perifollicular hemorrhage, corkscrew hairs, and petechiae that can evolve into ecchymoses and palpable purpura. Other symptoms include non-specific dyspnea, myalgias, and gingivitis. Anemia is multifactorial due to impaired iron absorption, bleeding, and commonly concurrent folate deficiency. Cardiovascular symptoms include hypotension, myocyte hypertrophy, EKG abnormalities, and even hemopericardium progressing to tamponade. If left untreated, scurvy is inevitably fatal due to infection or sudden cardiovascular collapse [5].

As seen in our case, the diagnosis of scurvy is clinical and confirmed with laboratory data. Risk factors include alcoholism, obesity, institutionalized elderly, concomitant gastrointestinal disease, poor dentition, and poor diet. One increasingly important risk factor is homelessness [6]. Compromised nutrition intake is related to being unhoused, and often compounded by alcohol or polysubstance abuse [7]. Early clinical recognition of these factors are important as confirmatory levels of ascorbic acid may take days to result. Additionally, treatment improves clinical outcomes and avoids unnecessary testing. The differential diagnosis is extensive, and common misdiagnoses include leukocytoclastic vasculitis, plasma cell dyscrasias, and ulcerative gingivitis [8]. Plasma ascorbic acid levels less than 0.2 mg/dL are diagnostic, although it should be noted plasma levels poorly reflect systemic tissue levels [2]. A leukocyte ascorbic level better measures overall body stores, but is not widely available.

Treatment of vitamin C deficiency leads to quick and remarkable improvement [9]. Acute management typically consists of 1 to 2 grams of supplementation daily for the first 3 days before tapering to 500 mg daily for the next week, then 100 mg for the next three months. Average stores of 1500 mg will last one to three months before onset of symptoms, which typically start when levels drop below 300 mg. Prognosis is excellent, as non-specific symptoms can resolve within five days and physical exam findings will often improve within 2 weeks. Complete recovery is often seen within 3 months [10].

DATA AVAILABILITY

All data generated or analyzed in this study are included in this article. Access to data is possible with permission from the responsible author.

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AUTHOR CONTRIBUTIONS

Writing the initial draft of the manuscript: BS, EB, JK, TS, Conceptualization and supervision: JC, TJ, TS, Medical management of the case: EB, JK, JC, Revising the manuscript critically and literature review: SC, BS, EB, TJ, TS.

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COMPETING INTERESTS

The authors declare no competing interests.

ETHICS APPROVAL

The project did not meet the definition of human subject research under the purview of the IRB according to federal regulations and therefore was exempt.

ADDITIONAL INFORMATION

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